$$\Rightarrow \quad \chi^2 = \sum_{i=1}^{N} \left(\frac{y_i - y(x_i, a, b)}{\sigma_i} \right)^2$$

The optimization condition implies

$$\frac{\partial \chi}{\partial a} = \frac{\partial \chi}{\partial b} = 0$$

Using $y(x_i, a, b) = a + x_i b$

$$\Rightarrow \quad \frac{\partial \chi}{\partial a} = \frac{1}{2\chi} \frac{\partial \chi^2}{\partial a} = 0$$

$$\Rightarrow \quad \frac{\partial \chi^2}{\partial a} = 0$$

$$\frac{\partial \chi^2}{\partial a} = \sum_{i=1}^{N} 2 \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) = 0$$

$$\frac{\partial \chi^2}{\partial b} = \sum_{i=1}^{N} 2 \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) x_i = 0$$

We can factor out 2 so we get conditions

$$\sum_{i=1}^{N} \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) = 0$$

$$\sum_{i=1}^{N} \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) x_i = 0$$

We may expand the sums as

$$\sum \frac{y_i}{\sigma_i^2} + \sum \frac{-a}{\sigma_i^2} + \sum \frac{-bx_i}{\sigma_i^2} = 0$$

As a & b are constants

$$\Rightarrow -a\sum \frac{1}{\sigma_i^2} - b\sum \frac{x_i}{\sigma_i^2} + \sum \frac{y_i}{\sigma_i^2} = 0$$

From the second equation

$$-a\sum \frac{x_i}{\sigma_i^2} - b\sum \frac{x_i^2}{\sigma_i^2} + \sum \frac{y_i x_i}{\sigma_i^2} = 0$$

As x_i, y_i & σ_i are know values we can calculate a & b

denote $\sum \frac{1}{\sigma_i^2} = \sigma$; $\sum \frac{x_i}{\sigma_i^2} = x$; $\sum \frac{y_i}{\sigma_i^2} = y$, $\sum \frac{x_i^2}{\sigma_i^2} = x^2$; $\sum \frac{x_i y_i}{\sigma_i^2} = xy$

$$\Rightarrow \left.\begin{aligned} y &= a\sigma + bx \\ yx &= ax + bx^2 \end{aligned}\right\} \quad \begin{aligned} a &= \frac{y - bx}{\sigma} \\ a &= \frac{yx - bx^2}{x} \end{aligned}$$

$$\Rightarrow \frac{y - bx}{\sigma} = \frac{yx - bx^2}{x}$$

$$\Rightarrow \cancel{yx}\ (y)(x) - (bx)x = \sigma yx - \sigma bx^2$$

$$\Rightarrow (y)(x) - \sigma yx = (bx)x - \sigma bx^2 = b((x)(x) - \sigma x^2)$$

$$\Rightarrow b = \frac{(y)(x) - \sigma yx}{(x)(x) - \sigma x^2} \quad //$$

$$\Rightarrow a = \frac{y - bx}{\sigma}$$

$$\Rightarrow a = \frac{1}{\sigma}\left(y - (x)\frac{(y)(x) - \sigma yx}{(x)(x) - \sigma x^2}\right)$$

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